

Wave 5.2R Stage 14 Cross-Formulation Forward Tournament

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Decision

Stage 14 audited eleven formulation groups against all six tournament entry requirements. H04, the bounded PF-A core-coefficient correction from Stage 5, is the only eligible entrant.

H04 is nominated for Stage 15 official forward verification, not accepted or promoted. The accepted periodic GRU `Fw` remains the incumbent. Registries are unchanged.

Tournament Contract

- Dataset: `polished_dataset`
- Input mode: setpoints
- Surface: `Fw`
- Split signature:
`c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16`
- Audited formulation groups: 11
- Eligible entrants: 1
- Frozen references: PF-A, accepted harmonic MLP `Fw`, accepted GRU `Fw`
- Training executed: none
- Official acceptance authorized: no

An entrant must pass every requirement:

- isolated gate;
- three-seed evaluation;
- matched-control improvement;
- leakage and causality checks;
- complete full-curve payload;
- inspectable inference path.

Missing evidence is a failed requirement. Synthetic certification cannot substitute for real-data qualification.

Entry Eligibility

Formulation	Isolated gate	Three seeds	Matched control	Eligible
Stage 4 hybrids	fail	no	fail	no
Stage 5 H04	pass	yes	pass	yes
Stage 5 H08	fail	no	raw only	no

Formulation	Isolated gate	Three seeds	Matched control	Eligible
Stage 6 spectral/Sobolev	fail	no	fail	no
Stage 7 mean/shape heads	fail	no	fail	no
Stage 8 compliance	fail	no	fail	no
Stage 9 K01	fail	no	partial	no
Stage 10 sparse laws	fail	no	fail	no
Stage 11 trust	fail	no	fail	no
Stage 12 optimization	fail	no	fail	no
Stage 13 weak oracle	synthetic only	no	no	no

Formulation	Causal/leakage	Full curves	Inspectable
Stage 4 hybrids	pass	yes	yes
Stage 5 H04	pass	yes	yes
Stage 5 H08	pass	yes	yes
Stage 6 spectral/Sobolev	pass	yes	yes
Stage 7 mean/shape heads	pass	yes	yes
Stage 8 compliance	pass	yes	yes
Stage 9 K01	pass	yes	yes
Stage 10 sparse laws	pass	yes	yes
Stage 11 trust	pass	yes	yes
Stage 12 optimization	pass	yes	yes
Stage 13 weak oracle	pass	no real payload	yes

Why H04 Enters

H04 passed all ten Stage 5 gates. Across seeds 314159 , 271828 , and 161803 , its raw MAE values are:

- 0.001725884 deg ;
- 0.001716240 deg ;
- 0.001805115 deg .

The mean is 0.001749080 deg and the standard deviation is 0.000039818 deg . Every seed beats PF-A and its matched direct-coefficient control. Runtime measured and target-derived inputs are absent, the complete 97 x 2048 test payload is preserved, and the coefficient correction and harmonic reconstruction path are explicit.

Why K01 Does Not Enter

K01 is a useful research component and improves H04 strongly on several scalar and curve metrics. It nevertheless failed the complete Stage 9 closure, P95, and declared chunk-equivalence gate. Conditional stability was therefore skipped. Stage 12 did not repair those failures. Scientific interest is not a substitute for the Stage 14 entry contract.

Why The Stage 13 Weak Residual Does Not Enter

The weak residual is implementation-valid and noise-robust on its analytical oracle, but it has no isolated real-data training result, no three-seed predictive evaluation, and no measured full-curve candidate payload. It remains available for future matched testing only.

Compatible Metric Surface

Model	Role	Raw MAE	Centered MAE	Offset error
H04	eligible entrant	0.001726	0.001356	0.000884
PF-A	analytical reference	0.001807	0.001385	0.000965
Harmonic MLP Fw	accepted reference	0.001694	0.001390	0.000825
Sequence GRU Fw	incumbent reference	0.001618	0.001382	0.000690

H04 improves PF-A by

- 4.49% on the frozen Stage 0 raw-MAE value;
- 2.11% on centered MAE;
- 8.36% on offset error.

Against the accepted GRU, H04:

- regresses raw MAE by 6.67% ;
- improves centered MAE by 1.92% ;
- regresses offset error by 28.17% .

This is a real tradeoff, not a scalar victory. H04 supplies the best compatible centered-shape value, while the GRU remains substantially stronger on raw and offset behavior.

Category Decisions

Category	Entrant result	Reference result	Decision
Raw error	H04 is sole entrant	GRU better	retain incumbent
Centered shape	H04 best	H04 beats all references	H04 advantage
Offset	H04 beats PF-A	GRU better	retain incumbent
Harmonic fidelity	H04 metrics available	contracts use unlike units	defer ranking
Robustness	H04 passes three seeds	references not seed-matched here	certify H04 only
Interpretability	bounded explicit coefficients	PF-A is white box	H04 inspectable
TwinCAT readiness	compact graph, no parity yet	PF-A parity exists	Stage 15 required
Balanced recommendation	H04 only entrant	GRU incumbent	nominate H04 for verification

The Stage 5 harmonic metrics use degree amplitude error and radian phase error. The neural Stage 0 references use percentage amplitude error and degree phase error. Ranking those values directly would be invalid. Stage 15 must regenerate all candidates on one official metric contract.

Interpretability And Deployment

H04 has an inspectable inference path

1. normalize speed, torque, and oil temperature;
2. evaluate the PF-A coefficient surface;
3. infer nine bounded complex coefficient corrections;
4. add corrections to the PF-A anchor;
5. reconstruct the periodic curve from explicit harmonic orders.

This is favorable for TwinCAT because intermediate coefficients and bounds can be exposed. It is not deployment-ready yet. The learned correction network has no accepted ONNX export, Python/ONNX parity, TwinCAT graph, or PLC parity evidence.

Tournament Decision

Stage 14 closes as a single-entrant tournament with frozen references:

- eligible entrant: H04;
- balanced Stage 15 nominee: H04;
- incumbent accepted forward model: periodic GRU F_w ;
- official acceptance: no;
- registry update: no;
- harmonic winner: unresolved pending common metrics;
- Stage 15: authorized.

H04 advances only to a separate official forward verification and deployment preparation step. It must not replace the GRU unless Stage 15 curve-first evidence and numerical parity justify acceptance.

Stage 15 Requirements

Stage 15 must

1. prepare a separate forward-only TE Curve Verification Pipeline launcher;
2. compare H04, PF-A, harmonic MLP, and GRU on one official curve-first surface;
3. normalize raw, shape, offset, harmonic, robustness, and visual evidence;
4. define and export the H04 inference graph;
5. prove Python/ONNX and PLC parity;
6. update registries only after an explicit acceptance decision.

Reproducibility

The machine-readable closeout is under

`output/analysis/wave_5_2r/stage14_cross_formulation_forward_tournament/` . It contains the entry matrix, hashed evidence inventory, compatible metric comparison, category matrix, and final decision. The analysis script supports both build and validation-only modes.